

UNIT 01

INTRODUCTION TO COMPUTERS



Student Learning Outcomes:

After the completion of this unit students will be able to:

- Recognize various Information and Communication Technology (ICT) devices and their applications. Summarize the different developments in the history of Computers.
- Define and differentiate computer hardware and software.
- Identify and analyze (basic) hardware components of a computer system (e.g. Input, Output, processor, memory, and storage).

Introduction to the Unit:

This unit covers the different components of computers (processors, input/output devices, and storage devices) and their functionality, advantages, technological development, and uses in the field of communication, business, and manufacturing. This unit contributes in recognizing computer systems and various ICT devices; differentiating hardware and software and analyzing the importance, advantages, and uses of ICT devices.



1.1. What is computer?

A computer is an electronic data processing machine that accepts the input (data), processes it, and produces output (information).

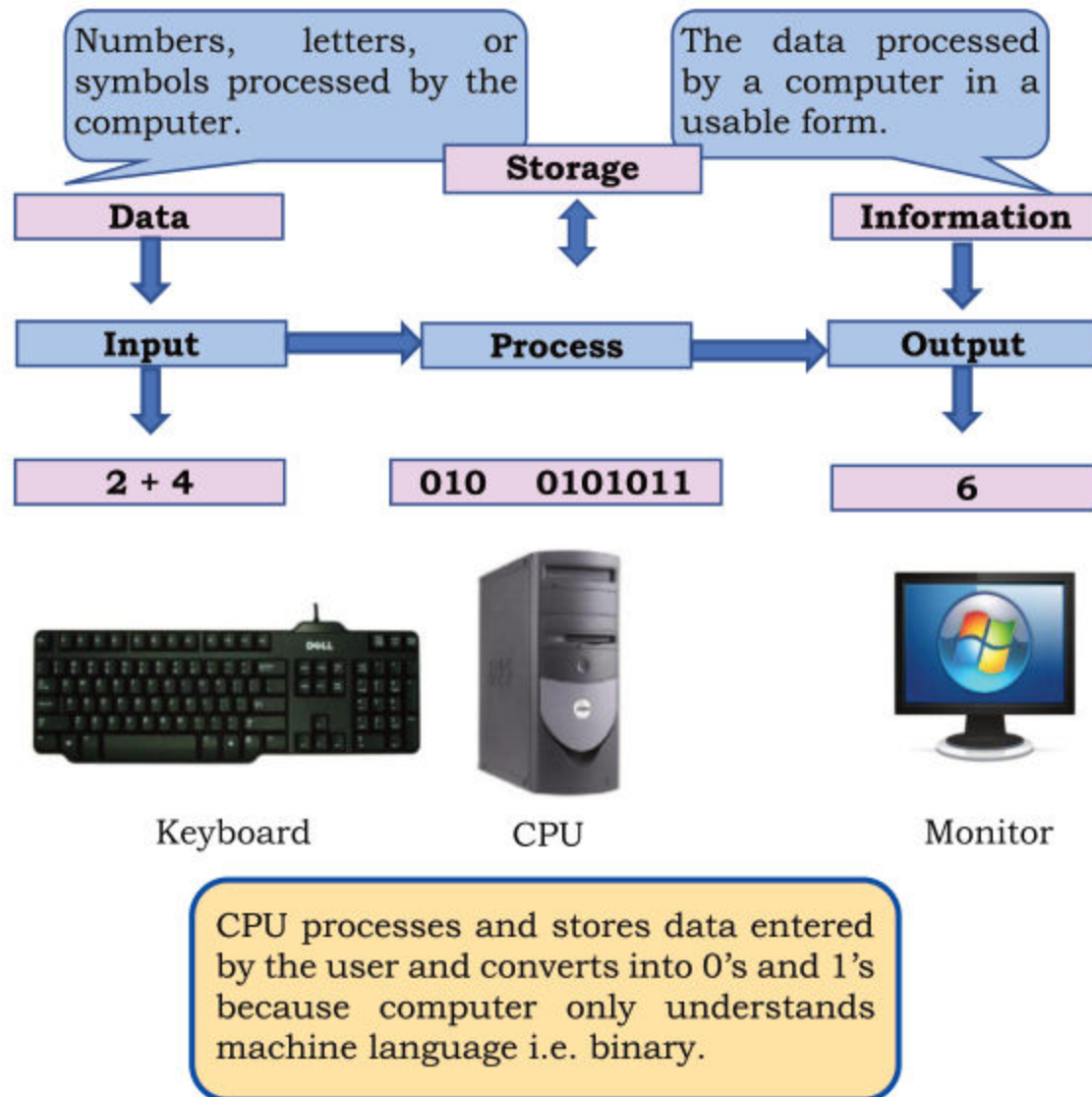


Figure 1.1 Computer

1.2. Difference between data and information

Data is raw, unorganized facts or symbols that represents quantities, measurements, or descriptions. It can be in the form of numbers, words, images, sounds, etc. Data has no context or meaning on their own. For example:

- For students in the class, data could be like names of students, their roll numbers, or their ages, etc.

S. No.	Name	Roll No.	Age
1	Abdullah	74	11y
2	Sahil	45	12y
3	Murkh	32	12y
4	Ali	27	13y

- For books, data could be the title of the book, subject, or author etc.

	Title	Subject	Author
1	Toot batoot	Poems	Sufi Tabasum
2	Sindh ki Kahani	History	Hamedda Khoro
3	A Christmas Carol	Stories	Charles Dickens
4	Chirion ki alif laila	Stories	Kirshan Chandar

Information is what we get when we organize and interpret data. It's processed data that has meaning, context and relevance. Information helps to answer questions, solve problems, or make decisions. For example:

- If we take the student's data and say, "Abdullah is younger than Ali," that's turns data into information.
- If we organize unarranged books (data) by subject or title, it becomes information. Now, we can search for any book easily.



Unorganized Books



Organized Books

Data



Information

	★	▲	♥
001	★	▲	♥
002	★	▲	♥
003	★	▲	♥
004	★	▲	♥
005	★	▲	♥
006	★	▲	♥
007	★	▲	♥

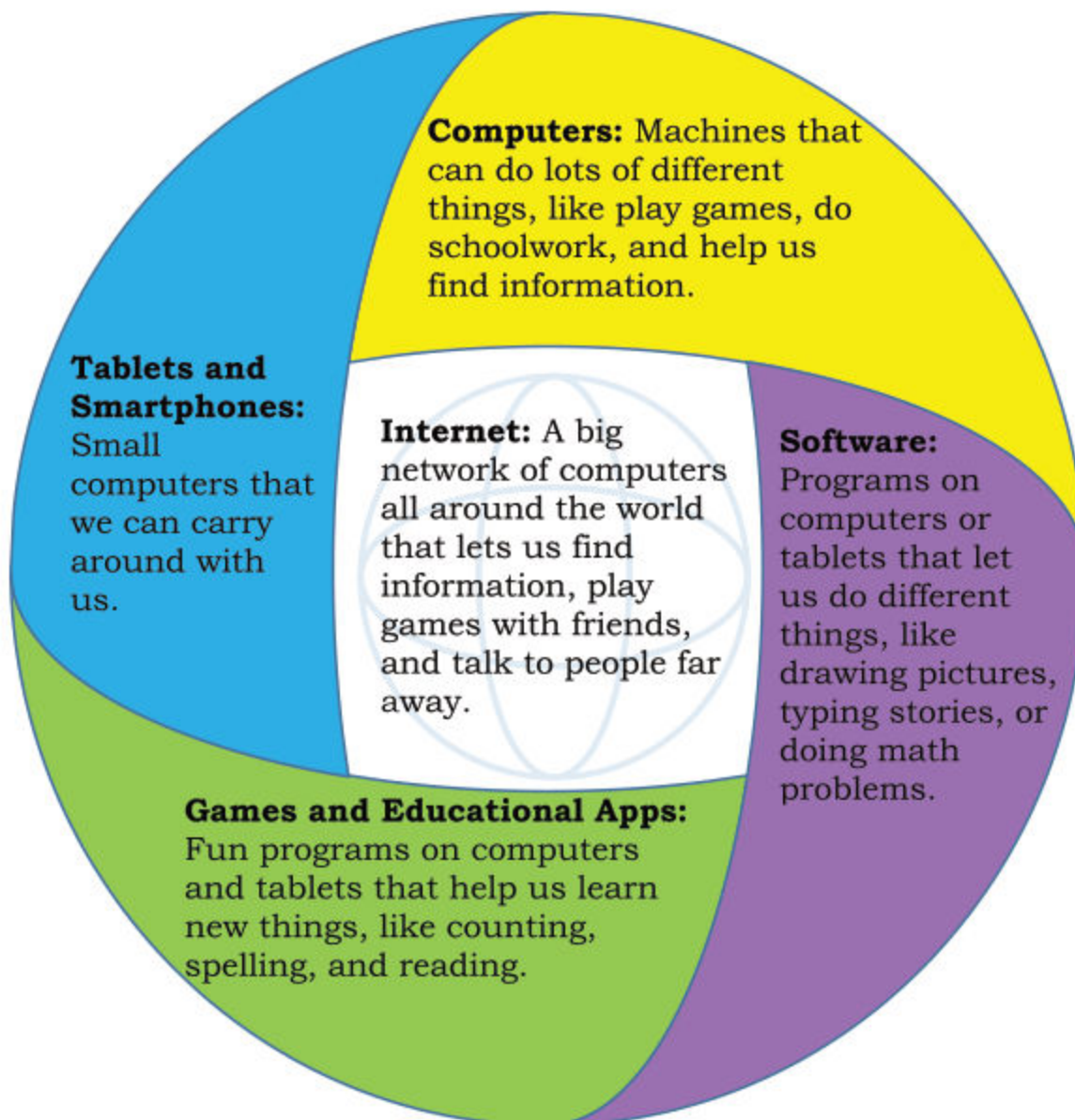
ACTIVITY

Use real-life examples from your environment to illustrate the concept:

Collect data (like counting objects in the classroom) and then work together to turn that data into useful information (like tabulating or graphing the results to see which object is most common).

1.3. Information and Communication Technology (ICT)

Information and Communication Technology (ICT) is all about using tools and resources to create, use, store, manage and communicate information. These tools include electronic devices like computers, tablets, and smartphones.



1.4. Evolution in the history of computers

The history of computers can be divided into five generations, based on major technological advancement. Each generation brought significant improvements in computing technology, leading to more powerful and versatile devices.

First Generation (1940s-1950s):

The first generation of computers used vacuum tubes for processing.

Introduction to Computers

These computers were large, expensive, and consumed a lot of power.



Vacuum Tube



ENIAC-1

Second Generation (1950s-1960s):

The second generation of computers used transistors instead of vacuum tubes, making them smaller, faster, and more reliable than the previous generation.



Transistor



CDC-164

Third Generation (1960s-1970s):

The third generation of computers used integrated circuits (ICs), which further reduced size, cost and power consumption while increasing speed and efficiency.



Integrated Circuit (IC)



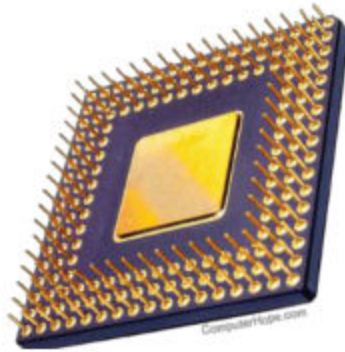
IBM Personal Computer

Fourth Generation (1970s-Present):

The fourth generation of computers used microprocessors, leading to the creation of personal computers (PCs).

The development of Graphical User Interface (GUI) made computers more user-friendly by allowing interaction through icons and windows.

The widespread adoption of the Internet in the 1990s revolutionized communication and access to information.



Microprocessor



Desktop Computer

Fifth Generation (Present and Future):

The fifth generation of computers is characterized by advancements in artificial intelligence (AI), quantum computing, Internet of Things (IoT), and other cutting-edge technologies.



Processor



Computing Devices

1.5. ICT Devices



Desktop Computer (CPU)



Printer



Scanner



Laptop



Tablet



Router



Satellite



Monitor



Biometric Scanners



Smart Board



Projector

1.6. Applications of ICT devices in daily life

Following are some explanations of use of ICT in various fields:

- **COMMUNICATION:**

Email: People can send messages quickly using computers and smartphones.



Video Calls: Applications like WhatsApp, Skype or Zoom let people see and talk to each other even if they are far away.



Social Media: Websites and apps like Facebook, Snapchat and Instagram let people share pictures and messages with their friends and family.



- **BUSINESS:**

Online Shopping: People can buy things from websites like Amazon and Daraz without visiting a store.



Digital Payments: People can pay for things using their phones or computers with apps like PayPal or Apple Pay.



Accounting Software: Businesses use software like QuickBooks to manage their financial records.



- **MANUFACTURING:**

Robots:

Robots help make things like cars and toys in factories.



Computer-Controlled Machines:

Computer control machines in factories to ensure accurate and efficient production.



Inventory Management: Computers help track the number of items in a factory or warehouse.



- **EDUCATION:**

Educational Games:

Students learn new concepts through fun games on computers, tablets, and smartphones.



Online Learning:

Websites like Khan Academy, Virtual University and W3schools offer lessons on various subjects.



Interactive Whiteboards:

Teachers use large screens connected to computers to show videos, play games, and draw pictures to help students for learning.



- **Research and Development (R&D):**

Research and development is exploring and applying new ideas to make things better.

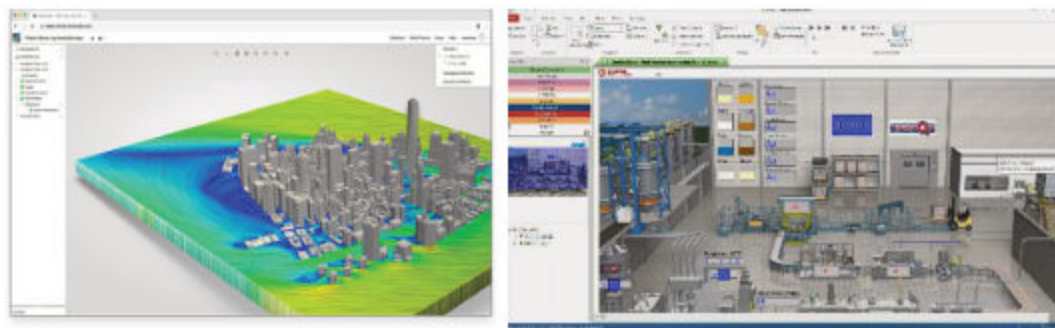
Example: Engineers build toys with new ideas for more fun.

ICT provide a large number of hardware and software tools for research and development.



- **Simulation Software:**

Computer programs are used to simulate experiments and test out ideas before building real **prototypes** e.g. Flight Simulator, Farm Frenzy, etc.



- **Data Analysis:**

Computers help researchers analyze large database to find patterns and make discoveries and predictions.



Collaboration Tools:

Tools like Google Docs and Dropbox enable teamwork, even from different locations.

These are just a few examples of how ICT is used in different fields to help people communicate, work, learn, and create new things.



1.7. Advantages and disadvantages of ICT devices

Advantages	Disadvantages
Enhanced Learning: ICT provides access to a wide range of educational resources, including interactive games, video sharing platforms, and apps like Khan Academy, Duolingo, BYJU'S, Ilmkidunya, Sabaq.pk, etc.	Distraction: ICT devices can be highly distracting for student, leading to decreased focus and productivity, if not used properly and monitored closely.
Communication: ICT helps to stay connected with people worldwide through email, messaging apps and social media like WhatsApp and Skype	Health Concerns: Excessive screen time on ICT devices can lead to various physical health issues, including eye strain, headaches, disrupted sleep patterns, and muscular injuries.

Efficiency and productivity:

Through ICT we can automate repetitive tasks and save a lot of time like Paint, Canva, Photoshop, YouTube, Spotify, etc.

Collaboration:

ICT devices facilitate collaboration among students by allowing them to work together on projects using tools like MS Teams, Zoom, Dropbox, etc.

Cybersecurity Risks:

ICT devices are vulnerable to cybersecurity threats such as viruses, malware, hacking, and phishing attacks.

Dependency:

Overreliance on ICT devices can lead to dependency and addiction.

1.8. Data Processing Cycle

Computers process data in a series of steps called the data processing cycle. Let's take a simple example of making bread ("Roti"), to illustrate the idea that data goes through different stages to become useful information.

Steps of the Data Processing Cycle

- **Input:** Like gathering your used clothes for laundry, at this stage data is collected and entered into the computer system.
- **Processing:** Like washing your clothes, the stage when computer works on the data to produce useful information.
- **Output:** Like collecting your washed clothes, output is where the processed information is presented to the user.
- **Storage:** Like storing the clothes in some cupboard, storing data on the computer's memory or hard drive.



1.9. Components of a Computer

A computer can be divided into two categories of components, **Hardware** and **Software**.

Hardware

Hardware refers to the physical parts of a computer that you can see or touch like Keyboard, Mouse and Display.



Software

Software consists of program (set of instructions) and data (consumed or produced by these programs) such as MS Windows, LINUX, Paint, Word, and the images or documents created through these programs.



Types of Software:

Software are divided into two broad categories:

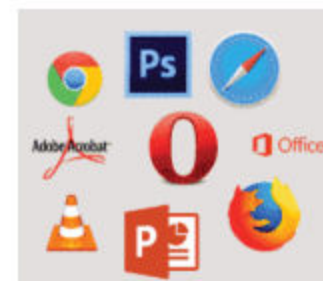
System Software:

This includes Operating System like Windows, LINUX, Android, Mac OS, etc. Device drivers like drivers for the sound card LAN (Local Area Network) card and VGA (Video Graphics Adapter). System software manages and controls the hardware resources of the computer.



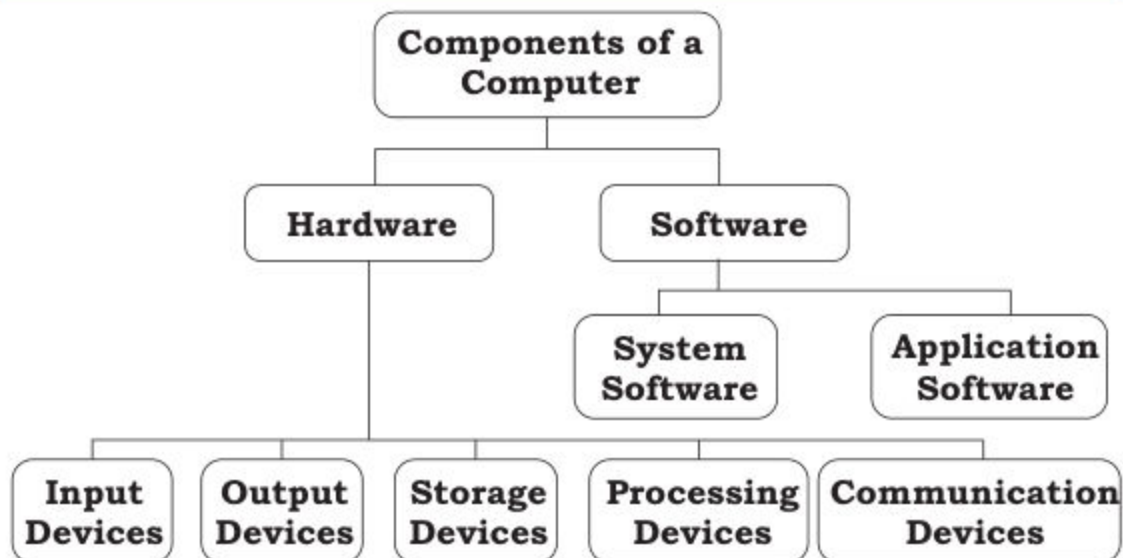
Application Software:

Application software are used to assist users in their daily routine tasks like word processing, image manipulation, games and entertainment, etc. like MS Word, MS Paint, Media Player, Chrome, Skype, etc.



ACTIVITY

Teacher will show and describe some software tools to students, also describe the purpose of using that particular tool, let them describe the advantages of using the software tool.



1.10. Hardware Components of a Computer

Hardware components of a computer can be categorized in following five broad categories as given in the above diagram.

1.10.1 Input Devices

Input devices are used to supply data from the user to the computer for processing.



	Keyboards are used to input textual data such as. Characters, Numbers, and Letters.
	A mouse inputs data by moving a cursor on the screen and clicking on Icons.

	Touch pad allows the user to move the cursor on the screen by moving the figure or stylus (a pen) on the pad. We can draw with drawing pad and play games with game pads
	Camera is used to input image and/ or video.
	Scanners are used to input image data. By scanning objects and store for further use.
	Microphones are used to input sound or voice data for further processing.
	Barcode and QR (Quick Response) scanners are used to input Barcode or QR code data for processing.
	Various kinds of detectors are used to capture quantities like light, fire, smoke, and motion for processing.

ACTIVITY

Teacher will demonstrate functions associated with various keys on the keyboard e.g. typing keys, control keys, function keys, navigation keys, and the numeric keypad.

1.10.2 Output Devices

After processing the data, the computer sends the information to the user through the output devices.



	Computer monitors are the primary output devices. The computer communicates textual, image, and video information through monitors.
	Printers are used to get textual and graphic information on paper.
	Audio or sound information are send through speakers.
	Projectors can be connected to the computers to get the textual, graphics, and video information on large screens.
	Plotters are used to print out large images on the papers and some other mediums like Pena flex and fabrics etc.
	3D printer print outs 3D objects with plastic, polymers, resins metal, and carbon fibers etc.



Fax are used to send or receive documents over the telephone lines

1.10.3 Processing Devices

Processing devices are components that help computers to do task. Example: Central Processing Unit helps computer to play music.

	All the processing and computation in a computer are performed by the microprocessor.
	ROM (Read Only Memory) is a small memory present in a computer. It contains the initial program BIOS (Basic Input Output System) to start the computer.
	RAM (Random Access Memory) is the main memory. All the programs and data for processing and the result produced from the processing reside in the RAM.
	Motherboard is the main board of a computer used to connect all the hardware like Power supply, Microprocessor, RAM, ROM, Storage devices, etc.

ACTIVITY

Teacher will Show the students some hardware components of computers physically and ask them to identify the components.

1.10.4 Storage Devices

Storage devices are used to store the data and information permanently.

	Hard Disk Drives (HDD) are the secondary storage of a computer where all programs and data are permanently stored.
	CDs DVDs and Blu-ray are removable storage, they can store 700 MB, 4.7GB, and 25 GB of data respectively.
	USB Flash Drives are designed to carry your digital stuff in your pocket.
	USB Flash cards (Memory Card) are designed to extend the available memory of various devices like cameras, cell phones, etc.

• Comparison of various storage devices

Storage Devices	Advantages	Disadvantages
Hard Drive	• Holds lots of data. • Usually, cheaper.	• Slower to read and write. • Can be big and heavy.
Solid State Drive	• Very fast to read and write data.	• More expensive.

	• Smaller and lighter.	• Holds less data than Hard Drives.
USB Flash Drive	• Small and portable. • Fast for transferring files.	• Can be lost easily. • Holds less data than Hard Drives and SSDs.
Cloud Storage	• Accessible from anywhere. • Can store large amounts of data.	• Requires internet connections. • Monthly fees may apply.

1.10.5 Communication Devices

Communication devices are used to build a network of ICT devices to communicate or exchange data.

	NIC (Network Interface CARD) or LAN (Local Area Network) cards are used to connect computer with other computers or communication devices.
	Network Switches are the devices to connect multiple computers and from a network.
	Routers are devices for connecting two or more networks. For example in our home, router connects our home network to the ISP's (Internet Service Provider) network.
	Bluetooth is used for exchanging data between fixed and mobile devices over short distances.

ACTIVITY

Teacher will Show the students some ICT devices and ask them to identify those devices.

COMPUTER MEMORY MEASUREMENT

The smallest unit of measuring information in computers is a “bit”. Which a binary digit and can either be “0” or “1”. Eight (8) bits makes a “byte”. Following table shows the various units of measurements.

Unit	Abbreviation	Storage Capacity
Bit	b	Single binary digit
Nibble	-	4 bits
Byte	B	8 bits
Kilo Byte	KB	1024 Byte
Mega Byte	MB	1024 KB
Giga Byte	GB	1024 MB
Terra Byte	TB	1024 GB

Summary

- **Computer** is an electronic data processing machine that accepts the input (data) and processes it to produce output (information).
- **Data** consist of raw, unorganized facts or symbols that represent quantities, measurements, or descriptions. Data doesn't have any context or meaning on their own.
- **Information** is what we get when we organize and interpret data. Information helps answer questions, solve problems, or make decisions.
- **Information and Communication Technology (ICT)** is all about using tools and resources to create, use, store, manage and communicate information.
- The development of computer throughout history can be divided into five generations based on the major technological

advancements. Each generation brought significant advancements in computing technology, leading to more powerful and versatile devices.

- **ICT** helps people in communication, business, manufacturing, education, and research & development.
- **ICT devices** have their advantages and disadvantages
- Computers process data in a series of steps called the data processing cycle. These steps are input, process, output, and storage.
- A computer can be divided into two categories of components, Hardware and Software.
- **Hardware** in the computer are the physical parts and Software is the program (set of instructions) and data (consumed or produced by these programs).
- **Software** is divided into two broad categories System software and Application software
- We categorize Hardware components of a computer into five broad categories Input Devices, Output Devices, Processing Devices, Storage Devices, and Communication Devices.

Important Terms To Remember

Vacuum Tubes: Early computers were made up of vacuum tubes. These were the first electronic devices used for computing. These were large and fragile glass tubes which consume much power and produce much heat when operated. Vacuum tubes were very slow.

Transistors: Transistors are discrete solid-state devices used in second generation computers. Transistors are fast, are small in size, consume less power, and produce less heat as compare to vacuum tubes.

Integrated Circuits (ICs): ICs are compact circuits composed of hundreds of transistors. One IC can replace hundreds of transistors resulting in more power and space saving, produce less heat and much faster than transistors.

Microprocessors: Microprocessors are equivalent to thousands of ICs on a single chip. Microprocessors are very small in size as compare to the functionality performed. Microprocessors are much faster than ICs, consume less power and produce very little heat.

AI: Artificial Intelligence (AI) is making computer to learn and make decision by their own selves.

IoT: Internet of Things (IoT) when everyday things, like lights, home appliances, and any physical devices, etc. can connect to the internet and share information to make our lives easier and more fun.

Simulation Software: Simulation is a class of software and applications that test and execute an experiment on computer (not in actual) and produces the result. In this way, we come to know the results without actually performing the test or experiment.

Data Analysis: Data Analysis tools are used to study very large amount of data on computers and make decisions.

Collaboration Tools: Collaboration Tools are used by many people working on same project from different locations. These peoples communicate with each other and share their work by using ICT and do not need to be on same place.

HDD: Hard Disk Drive, secondary storage device, stores data and applications magnetically.

SSD: Solid State Drive, secondary storage device, stores data and applications on semiconductor medium.

Device Drivers: Device Drivers are system software that controls and manages a particular devices like sound card, LAN card, or VGA card etc. attached to your computer. The operating system communicates with the device through its driver.

RAM: RAM is the primary memory of the computer. It stores data and programs during the execution of the program. RAM stores its contents until power is on.

ROM: ROM is the permanent memory of the computer, we cannot alter or change the contents of the ROM. The contents of the ROM are available after the power is off.

Word Processors: Word Processors are application software, used to create, edit, print, and store documents in computers.

Dropbox: It is a cloud-based storage service that allows users to store, access and share files from any device connected to the internet.

Prototype: A prototype is an initial model of a product or system used to test a concept or process.

Exercise

1. Encircle the correct answer.

- i. Select the storage device from the following options
 - a. Microprocessor
 - b. Memory card
 - c. Router
 - d. Motherboard
- ii. The following Generation of computers consumes more power
 - a. 1st generation
 - b. 2nd generation
 - c. 3rd generation
 - d. 4th generation
- iii. The more powerful and efficient generation of computer is the following
 - a. 1st generation
 - b. 2nd generation
 - c. 3rd generation
 - d. 4th generation
- iv. The device from the following options that can be used as both an input and output device is
 - a. Plotter
 - b. Monitor
 - c. Interactive board
 - d. Track ball
- v. The device that can store more data is
 - a. CD
 - b. DVD
 - c. Blu-ray disk
 - d. Floppy disk

2. Fill in the blanks with appropriate words given below.

Network switch, Output devices, System software, Application software, Router.

- i. The communication device is _____.
- ii. The class of devices used to get information from a computer is _____.
- iii. The device used to communicate data among networks is _____.
- iv. The type of software which is used to manage and control the computer itself is _____.
- v. The class of software used to assist the user in daily routine tasks is _____.

3. Provide the descriptive answers of the following questions.

- i. What is a computer?
- ii. How can we differentiate computer hardware from computer software?
- iii. Differentiate input devices from output devices.
- iv. List five Storage devices.
- v. Describe Processing devices.
- vi. What are the main features of various computer generations?
- vii. What are the applications of ICT in Education?
- viii. What are the applications of ICT in Industry?
- ix. How can a student get benefits from ICT?
- x. What are the disadvantages of ICT for young people?

4. Describe the Function of the following ICT devices:

i. Router	ii. LAN Card	iii. HDD
iv. Touch Pad	v. Head Phones	vi. Projector
vii. Microprocessor	viii. Mother Board	ix. Flash Drive
x. QR Code Scanner		

5. Name the devices:



A.



B.



C.



D.



E.



F.



G.



H.



I.

6. Label the following as either Input, Output, Storage, Processing, or Communication devices:



A.



B.



C.



D.



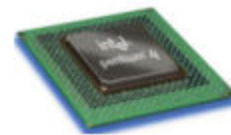
E.



F.



G.



H.



I.



J.

7. Group Activities

- i. Ask the students to draw some charts of computer hardware on drawing sheets.
- ii. Assign students a topic from ICT and ask them to prepare a small poster presentations (on drawing sheets).



Instructions for Teachers

- Teachers should engage students in interactive and hands-on activities.
- By keeping explanations simple and using relatable examples, students can easily grasp the difference between data and information.
- Introduce common ICT devices and tools such as computers, tablets, smartphones, keyboards, and other accessible devices.
- Use interactive demonstrations to show how ICT devices work.
- For example, demonstrate how a touchscreen device responds to touch or how a computer mouse moves the cursor on the screen.
- Show videos or animations that explain basic ICT concepts in a fun and engaging way. You may use some videos from YouTube.
- Provide hands-on activities where students can explore ICT devices and software under your supervision. For example: let them practice typing on a keyboard using educational typing games.
- Relate ICT concepts to students' everyday experiences. For example: discuss how they use technology at home or in school for communication (e.g., chatting with friends, sending emails) and entertainment (e.g., watching videos, playing games).
- Assign collaborative ICT projects where students work in small groups to create something using technology (if resources are available) or create some posters presenting the usage of ICT in their daily life (if computer resources are insufficient). For example: create a digital or paper-based storybook using text, images, and/or audio recordings.
- Provide feedback and guidance to students throughout their ICT activities. Assess their understanding through observation, discussions, and simple quizzes or worksheets.